ECON 0100 | Vignette A2 | Advantages | Solutions

Solution guide for the teaching team. Answers and working are in red; everything in black is what students see.

Colin Creevey can bake 20 cornish pasties (P) or 5 cauldron cakes (C) in one day. Katie Bell also bakes cornish pasties and cauldron cakes at a neighboring bakery. She can bake 15 pasties or 8 cakes in one day.

Q1 | Absolute Advantage

Set up a production table with both Colin and Katie's output per day. Who has the absolute advantage (AA) in pasties? In cakes?

AA in P: Colin

AA in C: Katie

Production per day	Pasties	Cakes
Colin	20	5
Katie	15	8

Absolute advantage is just "who makes more of it in a day": read straight down each column. Colin makes more pasties ($20 > 15$) and Katie makes more cakes ($8 > 5$).

Q2 | Comparative Advantage

Set up an opportunity cost table with Colin and Katie's opportunity costs for each good. Who has the comparative advantage (CA) in pasties? In cakes?

CA in P: Colin

CA in C: Katie

Each cell is "how much of the other good one unit costs," read off the production table: Colin's $20P = 5C$ gives $1C = 4P$ and $1P = \frac{1}{4}C$; Katie's $15P = 8C$ gives $1C = \frac{15}{8}P$ and $1P = \frac{8}{15}C$.

Opportunity cost	of 1 pasty	of 1 cake
Colin	$\frac{1}{4}$ C = 0.25 C	4 P
Katie	$\frac{8}{15}$ C $\approx$ 0.53 C	$\frac{15}{8}$ P = 1.875 P

Comparative advantage goes to the lower opportunity cost, again reading down each column. Pasties: $\frac{1}{4} < \frac{8}{15}$, so Colin. Cakes: $\frac{15}{8} < 4$, so Katie. Here AA and CA line up, which is why Q3 exists.

A useful check: the two entries in a row are reciprocals, so whoever has the lower cost in one good has the higher cost in the other. Nobody can have the comparative advantage in both.

Q3 | Better at Both

Suppose Katie buys a new oven and can now bake 25 pasties or 8 cakes in one day. Update both tables. Who has the comparative advantage in each good?

CA in P: Colin CA in C: Katie

Production per day	Pasties	Cakes
Colin	20	5
Katie (new oven)	25	8

Opportunity cost	of 1 pasty	of 1 cake
Colin	$\frac{1}{4}$ C = 0.25 C	4 P
Katie (new oven)	$\frac{8}{25}$ C = 0.32 C	$\frac{25}{8}$ P = 3.125 P

Katie now makes more of both goods, so she has the absolute advantage in both. Her opportunity costs change: $25P = 8C$ gives $1C = \frac{25}{8}P$ and $1P = \frac{8}{25}C$; Colin's row is unchanged. Pasties: $\frac{1}{4} < \frac{8}{25}$, so Colin still has the comparative advantage in pasties. Cakes: $\frac{25}{8} < 4$, so Katie keeps the comparative advantage in cakes.

The point of the question: Katie is better at both, but **relatively** much better at cakes ($\frac{8}{5}$ as many cakes as Colin, only $\frac{25}{20}$ as many pasties), so her cheap good is cakes and Colin's is pasties. Absolute advantage in both never means comparative advantage in both.